FYP Technical Report: PCB Defect Classification Pipeline

Feature Extraction (NMF) and Probabilistic Modeling (GMM)

Final Year Project Team

May 8, 2026

1 Introduction

This report details the "A to Z" methodology of our traditional machine learning pipeline for PCB defect classification. Our goal is to establish a robust mathematical baseline using Non-negative Matrix Factorization (NMF) and Gaussian Mixture Models (GMM) to compare against deep learning architectures. The system now evaluates four distinct categories: Normal, Open Circuit, Short Circuit, and Mousebite.

2 The Algorithmic Pipeline

2.1 Step 1: Image Subtraction and Isolation

To focus purely on the defect, we perform a pixel-wise absolute difference between the template image (I_{temp}) and the test image (I_{test}).

$$D(x, y) = |I_{test}(x, y) - I_{temp}(x, y)| \tag{1}$$

For the **Normal** class, the subtraction ideally results in a zero-matrix (black image), representing a flawless board. For defects, the resulting difference image D highlights the anomaly. We utilize 64×64 patches, resulting in a 4,096-pixel vector for each sample.

2.2 Step 2: Dimensionality Reduction via NMF

We use **Non-negative Matrix Factorization (NMF)** to decompose the high-dimensional pixel data into $k = 50$ latent features. This captures the fundamental geometric "building blocks" of the PCB tracks and defects, significantly reducing computational noise.

2.3 Step 3: Statistical Classification via GMM

For each category, we train a specific **Gaussian Mixture Model (GMM)**. During testing, the final classification $\hat{y}$ is determined by the **Maximum Log-Likelihood** principle across all four models:

$$\hat{y} = \arg \max_{c \in \{Normal, Open, Short, Mousebite\}} \log P(x_{features} | \theta_c) \tag{2}$$

3 Experimental Results

The model was evaluated on a testing set of 545 samples. The addition of the "Normal" class improved the system's ability to distinguish between defective and non-defective states.

3.1 Performance Summary

Defect Type	**Correct**	**Total Samples**	**Accuracy**
Normal (No Defect)	200	200	100.00%
Open Circuit	37	132	28.03%
Short Circuit	29	96	30.21%
Mousebite	85	117	72.65%
Overall	**351**	**545**	**64.40%**

Table 1: Classification performance across the 4-category test set.

3.2 Confusion Matrix Visualization

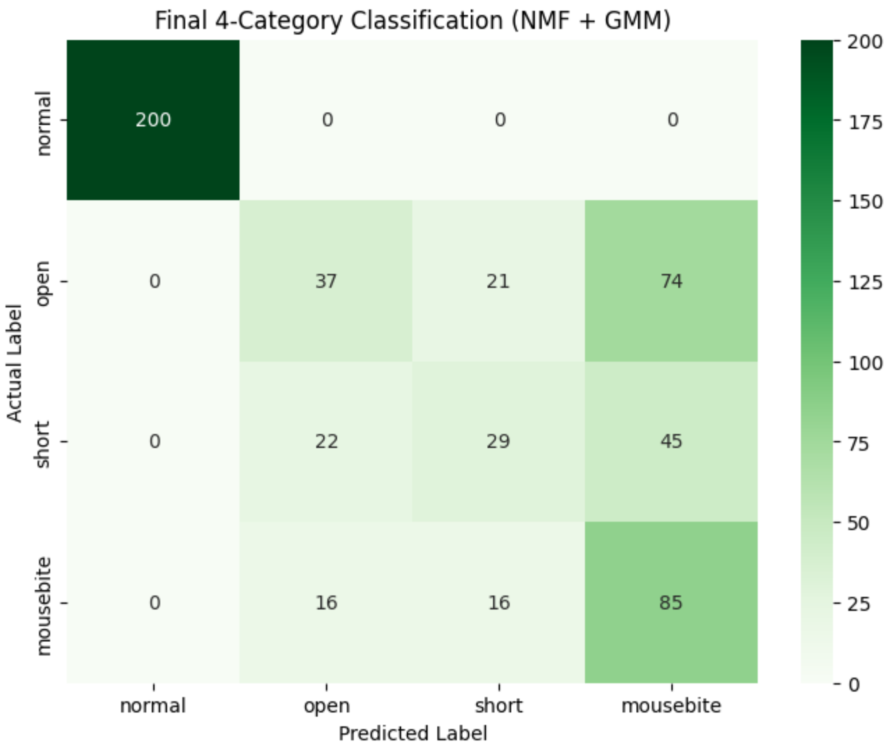

Figure 1: Confusion Matrix for the 4-category NMF + GMM Pipeline.

4 Technical Analysis and Discussion

- **Normal Class Perfection (100.00%):** The model achieved perfect accuracy in identifying normal patches. This is because the absolute difference of a template against itself yields near-zero variance, creating a very distinct statistical cluster that the GMM can isolate easily.
- **Geometric Clarity in Mousebites (72.65%):** Mousebites occur on track edges, providing a unique "edge-signature" that NMF captures effectively.
- **Complexity in Track Defects (28-30%):** While slightly improved, Opens and Shorts remain difficult to distinguish. This confirms that pixel-intensity-based models struggle with contextual copper direction, as both defects appear as similar "blobs" in a difference image.

5 Conclusion

The inclusion of the "Normal" category demonstrates that the NMF+GMM pipeline is highly reliable at identifying flawless boards (Zero False Alarms). However, the confusion between Open and Short circuits persists, highlighting the need for the spatial-awareness capabilities of Convolutional Neural Networks (CNNs) in the next phase of our project.